Data Mining & Machine Learning

Yong Zheng

Illinois Institute of Technology
Chicago, IL, 60616, USA

ILLINOIS TECH
College of Computing

Schedule: Before Midterm

- Data, Data Types
- Data Preprocessing
- Classification
  - KNN Classifier, Naïve Bayes Classifier, Decision Trees
  - SVM, Logistic Regression, ~~Neural Networks~~
  - Ensemble Classifier
  - Binary, Multi-Class and ~~Multi-Label classifications~~
- Clustering
  - K-Means, ~~Density-based Clustering~~, Hierarchical Clustering

Schedule: After Midterm

- Classification
  - Neural Networks, Multi-Label Classification
- Supervised Learning: Numerical Predictions
- Unsupervised Learning
  - Association Rules
  - Outlier Detections
- Java Tool for Data Science: Weka
- Semi-Supervised Learning
- Advanced Topics [Optional]
  - NLP, Search, Recommender Systems

Schedule

- Midterm Review
- Instructions for Final Project

Schedule

- Midterm Review
- Instructions for Final Project

Midterm Exam

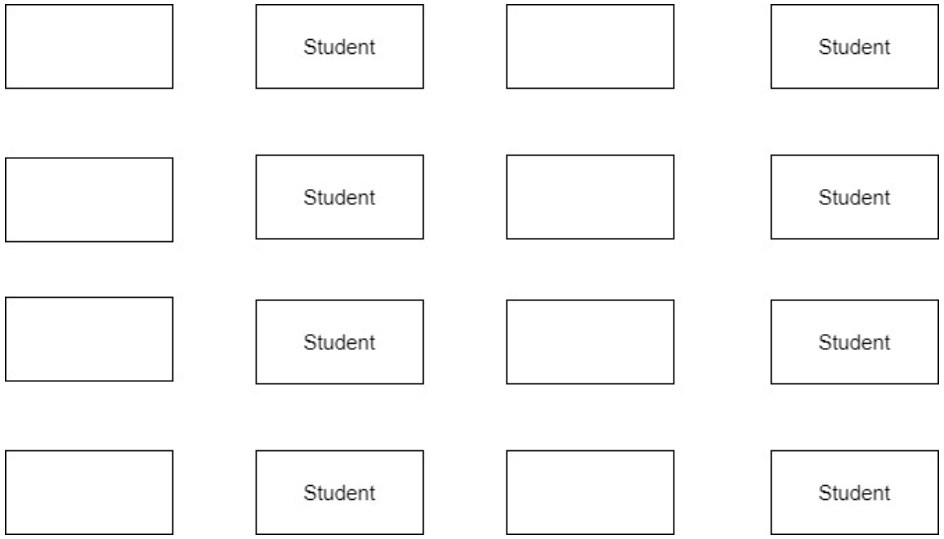
- Live Students take exams in main campus
- Remote students take exams in India/China
  - Contact me for details
- Online students take exams at selected locations
  - Rice Campus, IIT online will notify you the rooms
  - Main Campus
  - US cities, if we have IIT office there
  - MonitorEDU (you need to pay a fee, $13, for it)

Midterm Exam

- Time: Oct 22, 10:00 AM to 12:00 PM
- Location
  - Students will be split and distributed in 2 rooms RE-122, RE-124
  - Student Distribution
    - RE-122, for students with last name A to L
    - RE-124, for students with last name M to Z

Midterm Exam

- Students cannot seat together



Midterm Exam

- What you can NOT bring
  - Books, Computers, Laptops, notes
- What you can bring
  - Calculator [You cannot use Phone] [cannot be shared]
  - Pen
- Formula related to distance/similarities and entropy are given in the exam paper. No other formula included.
- Plagiarism
  - If you have plagiarism behaviors in exams, you got zero, and directly failed the class.

Knowledge Covered in Exams

- Data, Data Types
- Data Preprocessing
- Classification
  - KNN Classifier, Naïve Bayes Classifier, Decision Trees
  - SVM, Logistic Regression, ~~Neural Networks~~
  - Ensemble Classifier
  - Binary, Multi-Class and ~~Multi-Label classifications~~
- Clustering
  - K-Means, ~~Density-based Clustering~~, Hierarchical Clustering

Midterm Exam Review

- Type of Questions in exams
  - ~~Python Coding~~ => No such questions!
  - Concept Questions
    Examine whether you understand concepts/algorithms
    Scope: all knowledge covered (see previous page)
  - Problem Solving with Calculations
    Examine whether you can solve real-world problems
    Scope: similar to the questions with manual calculations we have in the assignments, or examples/practice given in slides

Knowledge Review

- Data, Data Types
- Data Preprocessing
- Classification
  - KNN Classifier, Naïve Bayes Classifier, Decision Trees
  - SVM, Logistic Regression, ~~Neural Networks~~
  - Ensemble Classifier
  - Binary, Multi-Class and ~~Multi-Label classifications~~
- Clustering
  - K-Means, ~~Density-based Clustering~~, Hierarchical Clustering
- ~~Numerical Predictions/Regressions~~

Knowledge Review

- Data, Data Types
- Data Preprocessing
- Classification
  - KNN Classifier, Naïve Bayes Classifier, Decision Trees
  - SVM, Logistic Regression, ~~Neural Networks~~
  - Ensemble Classifier
  - Binary, Multi-Class and ~~Multi-Label classifications~~
- Clustering
  - K-Means, ~~Density-based Clustering~~, Hierarchical Clustering
- ~~Numerical Predictions/Regressions~~

Knowledge Review

- Data, Data Types
  - Nominal
    - General nominal
    - Binary
    - Ordinal
  - Numerical
    - Discrete
    - Continuous

Knowledge Review

- Data, Data Types
- Data Preprocessing
- Classification
  - KNN Classifier, Naïve Bayes Classifier, Decision Trees
  - SVM, Logistic Regression, ~~Neural Networks~~
  - Ensemble Classifier
  - Binary, Multi-Class and ~~Multi-Label classifications~~
- Clustering
  - K-Means, ~~Density-based Clustering~~, Hierarchical Clustering
- ~~Numerical Predictions/Regressions~~

Knowledge Review

- Data Preprocessing
  - Reduce noise in numerical variables by smoothing
  - Reduce redundancy by correlation/dependency
  - Reduce dimensions by feature selection/reduction
  - Data normalization
  - Data discretization
    - Numerical to Nominal
    - Nominal to Numerical

Knowledge Review

- Data, Data Types
- Data Preprocessing
- Classification
  - KNN Classifier, Naïve Bayes Classifier, Decision Trees
  - SVM, Logistic Regression, ~~Neural Networks~~
  - Ensemble Classifier
  - Binary, Multi-Class and ~~Multi-Label classifications~~
- Clustering
  - K-Means, ~~Density-based Clustering~~, Hierarchical Clustering
- ~~Numerical Predictions/Regressions~~

Knowledge Review

- Classification
  - Supervised vs Unsupervised Learning
  - Classification: definition, tasks, evaluation, metrics
  - Classification Algorithms
    - KNN Classifier, Naïve Bayes Classifier, Decision Trees, SVM, Logistic Regression
    - Understand how each algorithm works, pros, cons
    - Know the specific requirements by each algorithm
    - Special issues, general issues and solutions
  - Ensemble Methods
  - Special issues: imbalance, micro/macro metrics, zero-probability issues, overfitting issues, etc.

Knowledge Review

- Data, Data Types
- Data Preprocessing
- Classification
  - KNN Classifier, Naïve Bayes Classifier, Decision Trees
  - SVM, Logistic Regression, ~~Neural Networks~~
  - Ensemble Classifier
  - Binary, Multi-Class and ~~Multi-Label classifications~~
- Clustering
  - K-Means, ~~Density-based Clustering~~, Hierarchical Clustering
- ~~Numerical Predictions/Regressions~~

Knowledge Review

- Clustering
  - Why it is unsupervised learning
  - Differences: partitional clustering, ~~density-based clustering~~, hierarchical clustering
  - Know how KMeans and hierarchical clustering work
  - Know how to evaluate clustering outputs

Schedule

- Midterm Review
- Instructions for Final Project

About Final Projects

- General Idea
- Requirements
- Where to find the data
- Steps to do

About Final Projects

- General Idea
- Requirements
- Where to find the data
- Steps to do

About Final Projects

- Goals by the Final Project

  Examine your practical skills

  Train your research and experimental capabilities

  Train your presentation skills

  Encourage you to learn going beyond the class

  Encourage you to solve problems by yourself

  Encourage you to work individually

  Encourage you to work in a team

About Final Projects

- Work In Team or Individually
  ~~You can choose to work individually~~
  You can work together in a team

  We have many students in this semester
  Team size: at least 3, at most 5
  Practical Tools
  You can only use Python in final projects

About Final Projects

- Grading Details
  I will grade your final projects, not TA
  It is different from grading assignments.
- In terms of grading assignments
  You get 100 as long as your answers are correct
- In terms of grading final projects
  Your project will be compared with other groups
  For example, you get 80. It does not mean your project is not good, but there are better ones
- At the end of your presentation, I will give you feedbacks

About Final Projects

- Grading Details
  - Presentation: Pending date
    We will run presentations over Zoom
    At least one team member must give presentation
    You will get feedbacks after presentations
  - Peer Evaluations
    Each team member will be asked to rate their teammates

About Final Projects

- Grading Details
  - Gradings
    - Team Grade, TG = score received for group projects
    - Individual Grade = scaled grade from peer evaluations
      Assume a group has 3 members, each member will be asked to rate their teammates

A	90
B	80
C	70

A	80
B	80
C	80

A	90
B	80
C	80

Average →

A	86.6
B	80
C	76.6

Assume TG for this group is 90.
scaled grade: A = 90, B = 83.1, C = 79.6

About Final Projects

- General Idea
- Requirements
- Where to find the data
- Steps to do

About Final Projects

- Requirements (The minimal requirements)
  - Data size is at least with 100K rows
  - Well-defined problems
    - What are the questions/problems to be solved in your data
    - What are the appropriate solutions and evaluation strategies
  - Appropriate solutions => a single solution is not enough, you need to compare multiple solutions
  - The degree of difficulty
    - It cannot be lower than the practice in the class or assignment
    - You should work on large data and void tiny data sets

About Final Projects

- Project Proposal
  - You need to submit a project proposal
    Proposal is a plan, you do not need to run any coding
    You can change your plan, if you change mind later
    - You must follow the proposal template
    - You must propose a feasible data, problem and solutions which meet the requirements in the last page
  - Proposal grade
    - Grade is binary =>>> 1 = approved, 0 = failed
    - We only validate whether your proposal makes sense. If so, you get score 1, otherwise you get zero, and you have to resubmit one

About Final Projects

- Different Data Mining Tasks
  - Classification/Clustering/Association Rules/Outlier
- Different Applications
  - Information Retrieval/Web Mining/Recommender Sys/ computer vision/Generative AI, etc.
- For all of these options, you must have clear evaluations and comparisons. You cannot run a single solution without comparing with others

About Final Projects

- Notes: how to find data and define research problems
  - Where to find the data? See the following slides
  - You must find a data you are interested in
  - You may need more than one data and integrate them together
  - You need to understand the data first, and then think about what are the problems you want to solve
  - Then figure out whether you can use the techniques or models you learnt to solve the problem
  - If the current data is not ideal, you need to find another data and go through the steps again until you are satisfied with one data

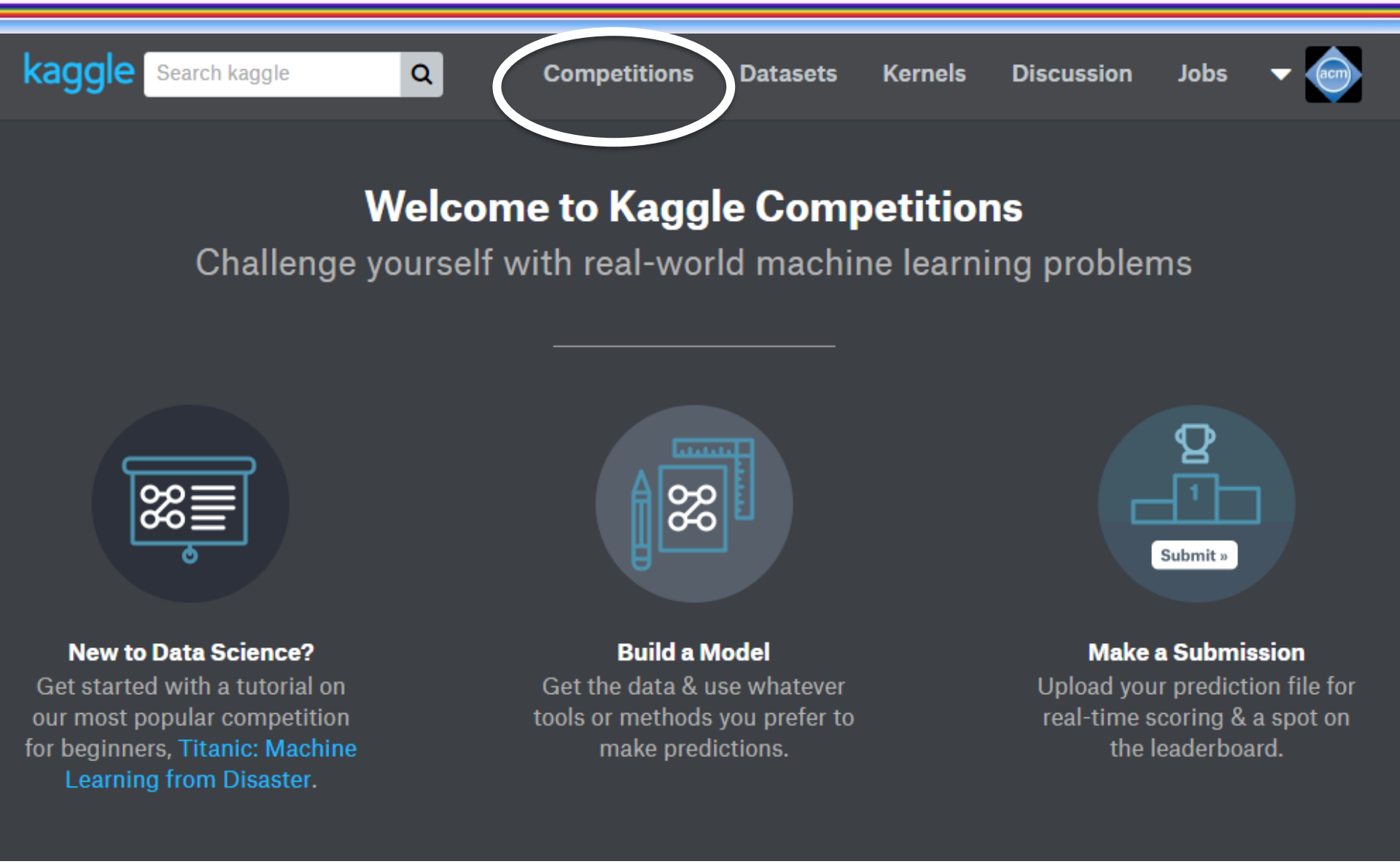
About Final Projects

- General Idea
- Requirements
- Where to find the data
- Steps to do

About Final Projects

- You should find data by yourself
  Kaggle: https://www.kaggle.com/
  Or other places to find appropriate data sets

Example: Data on Kaggle.com



Example: Data on Kaggle.com

kaggle Search kaggle Competitions Datasets Kernels Discussion Jobs

Datasets

Learn More New Dataset

445 featured datasets Sort By Hotness

Featured All Mine Upvoted

32

Pima Indians Diabetes Database

Predict the onset of diabetes based on diagnostic measures

UCI Machine Learning - updated 5 months ago

2,301 downloads
80 kernels
14 comments

30

Medical Appointment No Shows

Why do 30% of patients miss their scheduled appointments?

JoniHoppen - updated 20 days ago

1,344 downloads
64 kernels
13 comments

Raise some questions

- Do they make sense?
- Can we use data science techniques to solve them
- Can the proposed solutions solve the proposed problems?
- How to evaluate them?
- Any interpretations or explanations?
- How useful your outcomes are? Or, how can you use them in the future?

Rules

- You cannot use the following data
  - Weather data
  - Crime data
  - Housing data
  - Hotel data
  - Airline data
  - Game/Football/Basketball data
- They were frequently used by other students in previous years

About Final Projects

- General Idea
- Requirements
- Where to find the data
- Steps to do

About Final Projects

- ~~Step 1: fill in a survey (will be released later)~~
- Step 2: build the team
- Step 3: write your project proposal
- Step 4: once your proposal is qualified, you can start working. Otherwise, you need to revise and resubmit the proposal
- Step 5: final project presentations

About Final Projects

- Step 3: write the proposal
  - The template has been uploaded
  - Each team should ONLY submit one copy. For example, if there are 3 students in your team, it is enough for one student to submit the proposal. Do NOT submit multiple copies by different students!
  - Note: you may start working on it as soon as possible. Every semester, there are many teams whose proposal is disqualified.

About Final Projects

- Step 1: fill in a survey
- Step 2: decide to work individually or by team
- Step 3: write your project proposal
- Step 4: once your proposal is qualified, you can start working. Otherwise, you need to revise and resubmit the proposal
- Step 5: final project presentations

About Final Projects

- Step 5: final project presentations
  - Once you complete your presentation, I will give you feedbacks
  - Peer evaluations will be collected, so that individual grades can be derived from team grade

About Final Projects

- Proposal due: pending
  - Must be submitted by one member of a group
  - Only one member, just put the name/email of all group members in the proposal
- Presentation: pending